To enable reporting of events by carriers and tracking of events by the ordering party, the ordering party can create a freight document for tracking and send it to the SAP Business Network for Logistics system using the SOAP interface from Transportation Management (TM) on S/4HANA, `TransportationOrderGenericTrackedProcessRequest_Out`. For more information on the API, see Provider API to Create or Update Track Process.

> **Note**
>
> For road freight orders, if the shipper has enabled tracking via `TransportationOrderRequest_Out`, the reporting of events can be enabled automatically after the carrier confirms the freight order in the **Freight Orders for Confirmation** app. However, in this case, the carrier can only report events on the freight order manually in the **Manage Freight Execution** app.

2 - Send Freight Document for Tracking

In the SAP Business Network for Logistics system, the carrier receives the create or update tracking request and can view and manage the request in the Manage Freight Execution app. The carrier can create a tracking request in their carrier network using the `POST LbnOrderforTracking` service in the public REST API, Consumer API for Order Tracking (FC). The carrier can receive the real-time event status and predictive estimated time of arrival information from their carrier network.

3 - Receive Events and ETA

The carrier can send this information to their SAP Business Network for Logistics system using the Manage Freight Execution app. Alternatively, the carrier can use the Events service in the public REST API, Provider API for Order Events (FC) to report these events.

4 - Update Freight Document with Events

The ordering parties can view the status of the execution of their freight documents in the Track Freight Movement app. The ordering parties can view information sent by the carrier. The ordering party can also update the freight order with the event information in their external transportation system using the SOAP interface from TM on S/4HANA, `TransportationEventBulkNotification_In`. For more information on the API, see Consumer API to Notify Events.

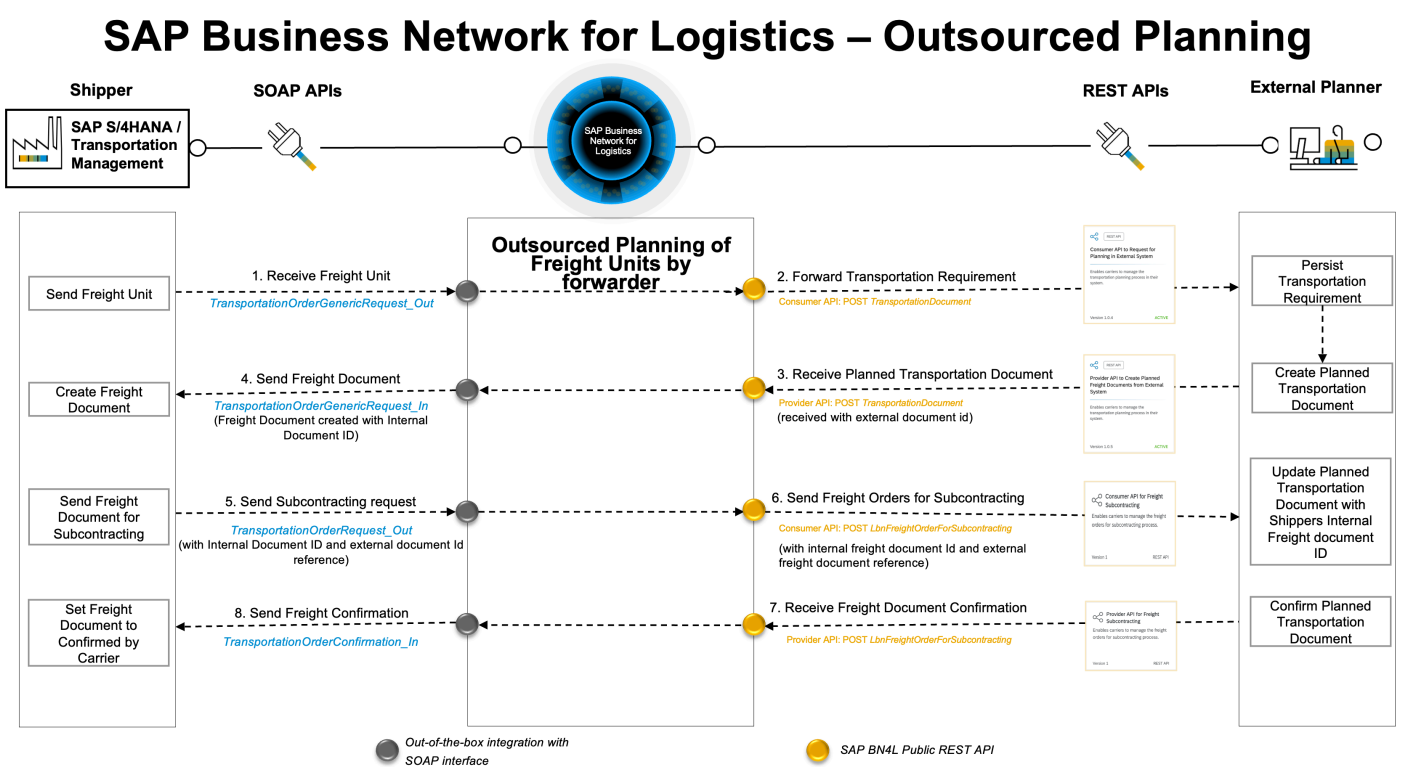
Post tracking, during the execution process, the execution status of the freight order is set as **In Execution**. On calculation of charges, the calculation status is set as **Calculated**. At this point, the freight order is sent to the carrier for settlement.

Outsourced Planning of Freight Units

The ordering party on Transportation Management (TM) on SAP S/4HANA system can create freight units and outsource the planning of freight units externally to a forwarder. SAP Business Network for Logistics has enabled the possibility of receiving such freight units from the ordering party and forwarding the same to the forwarder (onboarded on SAP Business Network for Logistics) using the public REST API. Post planning the freight units, the forwarder can send the planned freight document to SAP Business Network for Logistics using the public REST API which would be routed to the original ordering party's backend TM on S4HANA system. Post creation of the freight order or freight booking in the ordering party's S4HANA system, the freight document processing takes place.

In this end-to-end process, SAP Business Network for Logistics acts as the integration platform for the ordering party and forwarder.

Transportation Requirements Creation Scenario



1 - Receive Freight Unit

The ordering party creates transportation requirements from a sales order, purchase order, or delivery documents in their TM on SAP S/4HANA system or the external transportation system. The ordering party can send these transportation requirements or freight units to the carrier or forwarder and request the forwarder to perform planning activities in the carrier's system. To do so, the ordering party can use the SOAP interface from TM on S/4HANA, `TransportationOrderGenericRequest_Out`.

2 - Forward Transportation Requirement

The carrier receives the transportation requirements or freight units in the SAP Business Network for Logistics system. The carrier can receive and process the requirements using the `POST TransportationDocument` service in the public REST API, Consumer API to Request for Planning in External System.

3 - Receive Planned Transportation Document

Once the carrier creates a planned freight document in their system, they can send the planned transportation document such as the freight order or a freight booking containing the transportation requirements back to the ordering party. To do so, the carrier uses the `POST TransportationDocument` service in the public REST API, Provider API to Create Planned Freight Documents from External System.

4 - Send Freight Document

The ordering party receives the planned freight document from the carrier and updates their external system. To do so, the ordering party can use the SOAP interface from TM on S/4HANA, `TransportationOrderGenericRequest_In`.

Subsequent Subcontracting Process

5 - Send Subcontracting request

The ordering party creates a freight order from their external transportation system and sends it to the carrier using the SOAP interface from TM on S/4HANA, `TransportationOrderRequest_Out`. For more information on the API, see Provider API to Create Freight Order for Subcontracting.

The external ID for the planned freight document must be sent as a document reference with type code 1122 (Transportation Order).

6 - Send Freight Orders for Subcontracting

The carrier receives this freight order in the Freight Orders for Confirmation app. Alternatively, the carrier can also use the POST `LbnFreightOrderForSubcontracting` service in the public REST API, Consumer API for Freight Subcontracting to receive and process the freight order. The carrier can use this API to also send the freight order details to their carrier network.

The external ID for the planned freight document must be sent as a document reference with type code 1122 (Transportation Order).

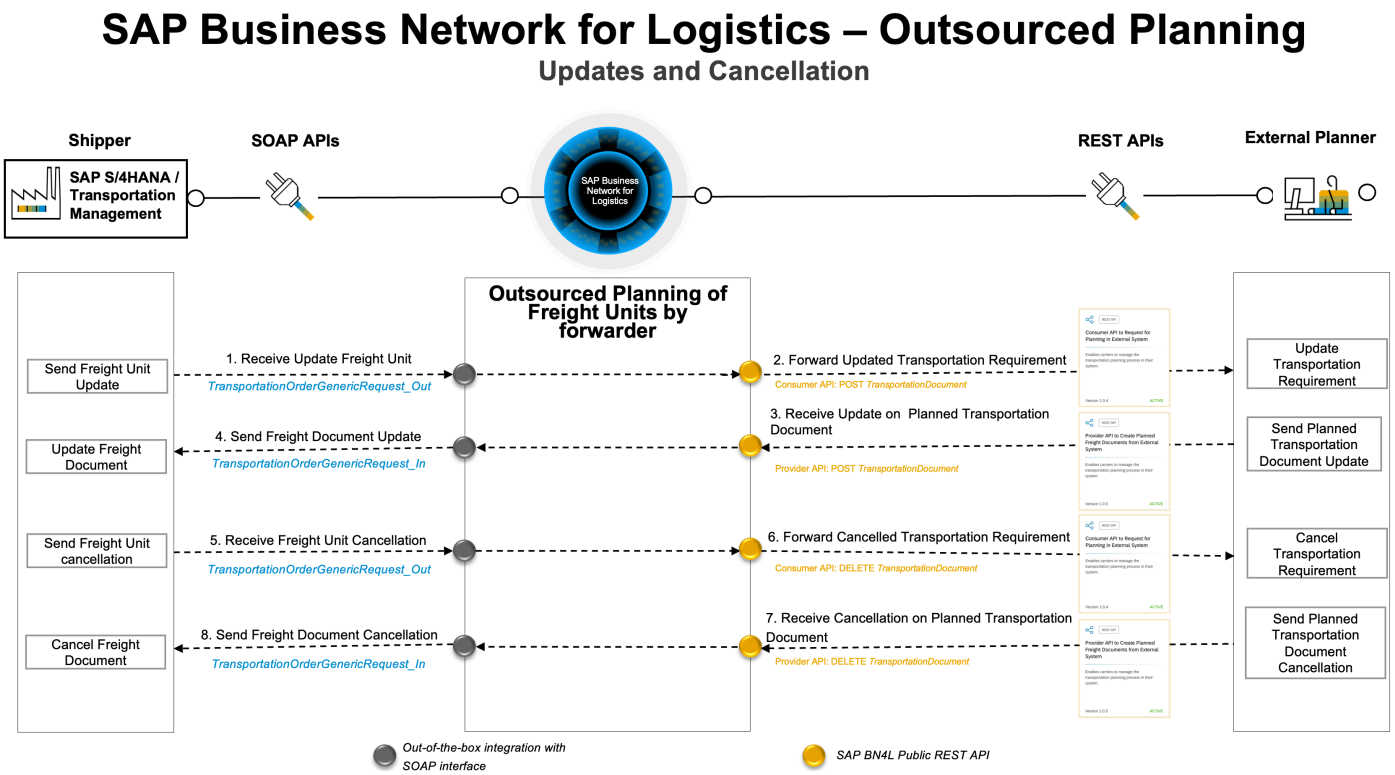
7 - Receive Freight Document Confirmation

The carrier responds to this freight order by confirming or rejecting the freight order in their SAP Business Network for Logistics system. The carrier can confirm or reject the freight order from the **Freight Orders for Confirmation** app. Alternatively, the carrier can also use the POST `LbnFreightOrderForSubcontracting` service in the public REST API, Provider API for Freight Subcontracting to confirm or reject the freight order and send the response to the ordering party.

8 - Send Freight Confirmation

The ordering party receives the carrier's response in their external transportation system using the SOAP interface from TM on S/4HANA, `TransportationOrderConfirmation_In`. For more information on the API, see Consumer API to Confirm Freight Order for Subcontracting.

Transportation Requirements Update Scenario



1 - Receive Updated Freight Unit

The ordering party can update the transportation requirements or updated freight units and send them to the carrier. To do so, the ordering party can use the SOAP interface from TM on S/4HANA, `TransportationOrderGenericRequest_Out`.

2 - Forward Updated Transportation Requirement

The carrier receives the updated transportation requirements or updated freight units in the SAP Business Network for Logistics system. The carrier can receive and process the updated requirements using the POST `TransportationDocument` service in the public REST API, Consumer API to Request for Planning in External System.

3 - Receive Update on Planned Transportation Document

Once the carrier updates the planned freight document in their system, they can send the document back to the ordering party. To do so, the carrier uses the `POST TransportationDocument` service in the public REST API, Provider API to Create Planned Freight Documents from External System.

4 - Send Freight Document

The ordering party receives the updated planned freight document from the carrier and updates their external system. To do so, the ordering party can use the SOAP interface from TM on S/4HANA, `TransportationOrderGenericRequest_In`.

Transportation Requirements Cancellation Scenario

5 - Receive Freight Unit Cancellation

The ordering party can send a cancellation request to the carrier to cancel the freight unit planning request. To do so, the ordering party uses the SOAP interface from TM on S/4HANA, `TransportationOrderGenericRequest_Out`.

6 - Forward Cancelled Transportation Requirement

The carrier receives the cancellation request from the ordering party and manages it using the `DELETE TransportationDocument` service in the public REST API, Consumer API to Request for Planning in External System.

7 - Receive Cancellation on Planned Transportation Document

Once the carrier cancels the planned freight document in their system, they can send the cancellation back to the ordering party. To do so, the carrier uses the `DELETE TransportationDocument` service in the public REST API, Provider API to Create Planned Freight Documents from External System.

8 - Send Freight Document Cancellation

The ordering party receives the cancellation of the planned freight document from the carrier and the freight document is cancelled in their external system. To do so, the ordering party can use the SOAP interface from TM on S/4HANA, `TransportationOrderGenericRequest_In`.

Note

- The planning in the external system is managed only using the API scenario.
- When a transportation requirement is cancelled by the ordering party, the carrier may cancel the associated planned freight document or update the planned freight document to remove the cancelled transportation requirement.

Freight Settlement

SAP Business Network for Logistics enables the carriers to settle the freight documents with their customers.

Note

In this document, the term **Shipper** refers to the company that has onboarded on SAP Business Network for Logistics. The term **Ordering Party** refers to the multiple purchasing organizations of the shipper in their backend system.

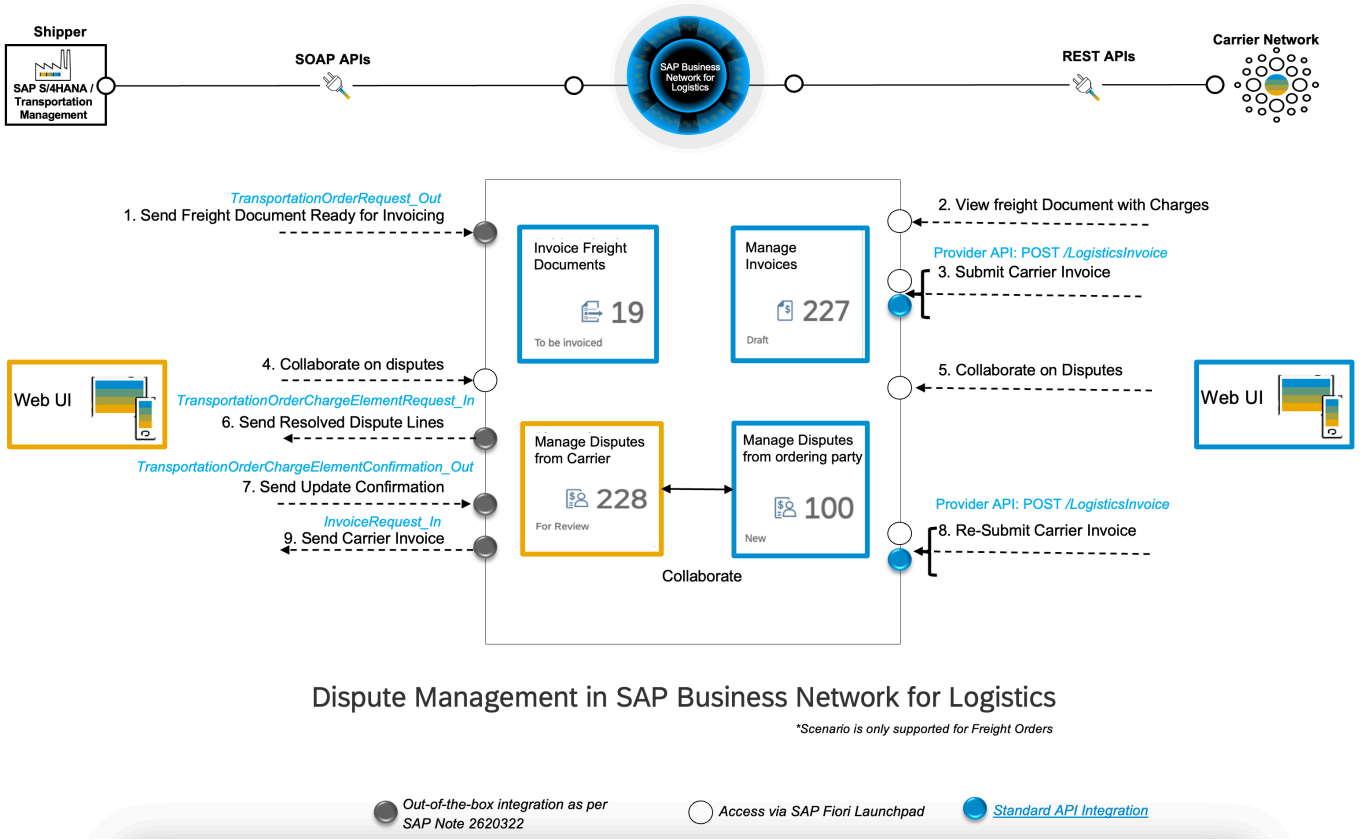
The carrier receives and processes the freight documents that are ready for settlement in the Invoice Freight Documents app and can create an invoice for a freight document from this app. The created invoice contains the planned charges sent by the ordering party. The created invoices can be further processed before submitting the invoice to the shipper in the Manage Invoices app.

In case of discrepancies, the system creates disputes, which can be collaboratively resolved by the shipper and the carrier using the **Manage Disputes** app.

Based on the system in which the shipper handles the dispute process, the settlement process differs. For more information, see Dispute Management in SAP Business Network for Logistics and Dispute Management in SAP TM on S/4HANA.

Dispute Management in SAP Business Network for Logistics

In this scenario, the ordering party handles the dispute process in SAP Business Network for Logistics.



Dispute Management in SAP Business Network for Logistics

**Scenario is only supported for Freight Orders*

1 – Send Freight Document Ready for Invoicing

The ordering party sends the freight document ready for invoicing that is, when the status of freight order is **In Execution** and the charges are calculated, from the Transportation Management (TM) on SAP S/4 HANA system to the SAP Business Network for Logistics system using the SOAP interface, `TransportationOrderRequest_Out`. For more information on the API, see Provider API to Create Freight Order for Settlement.

2 – View Freight Document with Charges

In the SAP Business Network for Logistics system, the carrier can receive and process the freight document with charges in the Invoice Freight Documents app. From the **Invoice Freight Documents** app, the carrier can create invoices and manage them in the Manage Invoices app.

3 – Submit Carrier Invoice

The carrier can manage and submit invoices in the **Manage Invoices** app. Alternatively, the carriers can also use the POST `LogisticsInvoice` service in the public REST API Provider API for Invoicing to submit invoices.

When the carrier submits an invoice, the system checks the invoice against the data contained in the freight document. If it detects inconsistencies between the invoice and the freight document, the system automatically creates a dispute. The shipper can view the disputes in the Manage Disputes app.

4, 5 – Collaborate on Disputes

The ordering party can then either accept the invoice or the ordering party and the carrier can engage in a dialog until the dispute is resolved.

The carrier can manage the disputes in the Manage Disputes app for carriers.

6 – Send Resolved Dispute Lines

Once the dispute is resolved, the freight document charges in TM on SAP S/4 HANA system are updated using the SOAP interface, `TransportationOrderChargeElementRequest_In`. For more information on the API, see Consumer API to Update Freight Order with Dispute Charges.

7 - Send Update Confirmation

Once the charges are updated, the system sends the confirmation to the carrier from their TM on SAP S/4 HANA system using the SOAP interface, `TransportationOrderChargeElementRequest_Out`. For more information on the API, see Provider API to Confirm Freight Order Dispute Charges.

8 - Re-Submit Carrier Invoice

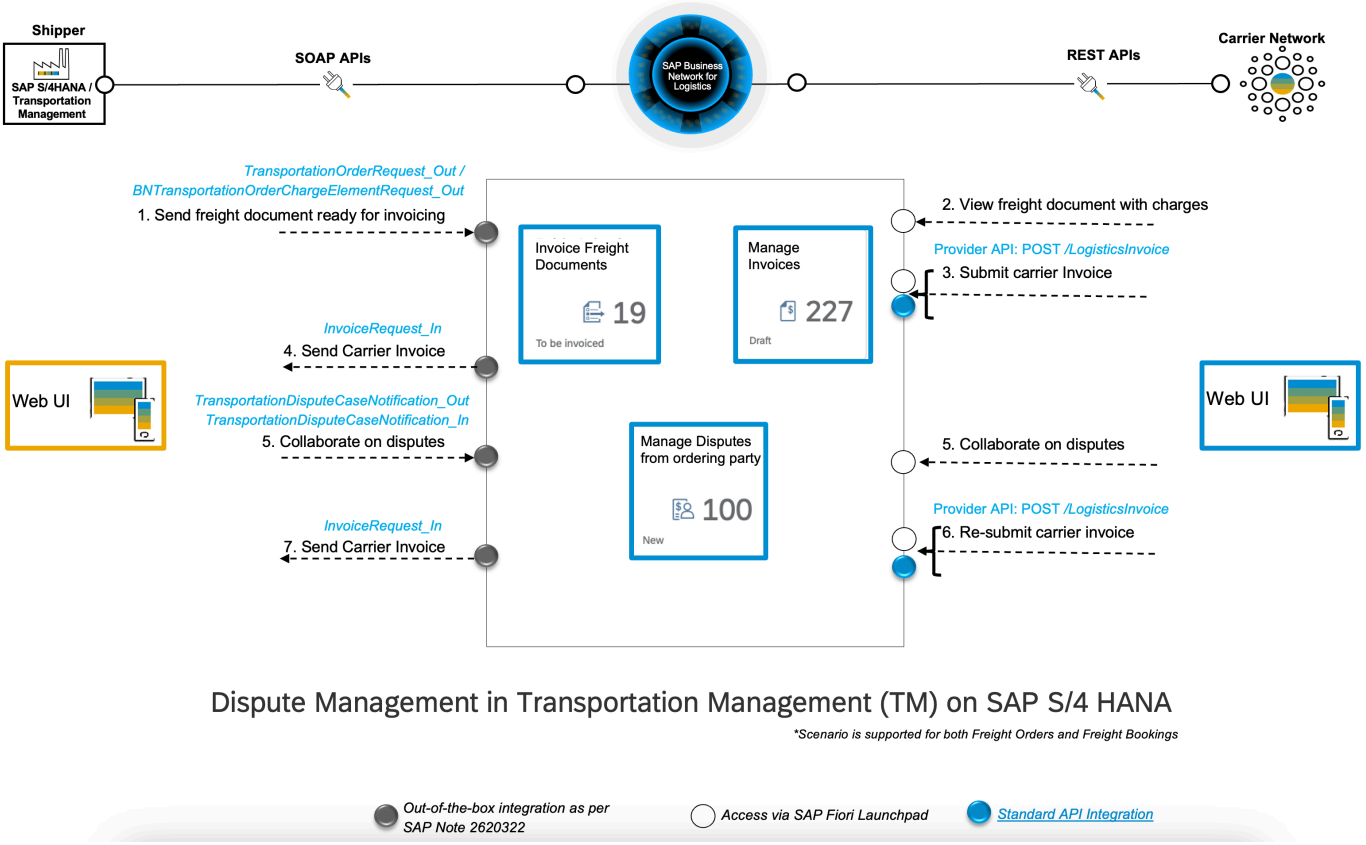
Once the confirmation is received, the dispute is set as **Resolved** and the carrier can re-submit the invoice with the updated charges using the **Manage Invoices** app. If the invoice is created via API, then the earlier invoice is rejected and the carrier must submit the new invoice with the updated charges using the `POST LogisticsInvoice`service in the public REST API, Provider API for Invoicing.

9 - Send Carrier Invoice

The submitted invoice is sent to the ordering party's TM on SAP S/4 HANA system using the SOAP interface, `InvoiceRequest_In`. For more information on the API, see Consumer API to Create Invoice.

Dispute Management in Transportation Management (TM) on SAP S/4 HANA

In this scenario, the ordering party handles the dispute process from their Transportation Management (TM) on SAP S/4 HANA system.



Dispute Management in Transportation Management (TM) on SAP S/4 HANA

*Scenario is supported for both Freight Orders and Freight Bookings

1 – Send Freight Document Ready for Invoicing

The ordering party sends the freight document ready for invoicing that is, when the status of freight order is **In Execution** and the charges are calculated, from the TM on SAP S/4 HANA system to the SAP Business Network for Logistics system using the SOAP interfaces, `TransportationOrderRequest_Out` or `BNTransportationOrderChargeElementRequest_Out`. For more information on the API, see Provider API to Create Freight Order for Settlement.

2 – View Freight Document with Charges

In the SAP Business Network for Logistics system, the carrier can receive and process the freight document with charges in the Invoice Freight Documents app. From the **Invoice Freight Documents** app, the carrier can create invoices and manage them in the Manage Invoices app.

3 – Submit Carrier Invoice

The carrier can manage and submit invoices in the **Manage Invoices** app. Alternatively, the carriers can also use the `POST LogisticsInvoice` service in the public REST API Provider API for Invoicing to submit invoices.

4 - Send Carrier Invoice

The submitted invoice is sent to the ordering party's TM on SAP S/4 HANA system using the SOAP interface, `InvoiceRequest_In`. For more information on the API, see Consumer API to Create Invoice.

When the carrier submits an invoice, the system checks the invoice against the data contained in the freight document. If it detects inconsistencies between the invoice and the freight document, the system automatically creates a dispute. The shipper can view the disputes in the Manage Disputes app.

5, 6 – Collaborate on Disputes

The ordering party can then either accept the dispute or the ordering party and the carrier can engage in a dialog until the dispute has been resolved.

The carrier can manage the disputes in the Manage Disputes app for carriers. The ordering party can manage the disputes in their TM on SAP S/4 HANA system and collaborate with the carrier using the SOAP interfaces, `TransportationDisputeCaseNotification_Out` and `TransportationDisputeCaseNotification_In`. For more information on the APIs, see the Provider API to Create or Update Dispute and Consumer API to Create or Update Dispute.

7 - Re-Submit Carrier Invoice

Once the dispute is resolved, the carrier can re-submit the invoice with the updated charges using the **Manage Invoices** app. Alternatively, the carrier can also submit a new invoice using the `POST LogisticsInvoice` service in the public REST API, Provider API for Invoicing to receive and re-submit the invoice.

8 - Send Carrier Invoice

The submitted invoice is sent to the ordering party's TM on SAP S/4 HANA system using the SOAP interface, `InvoiceRequest_In`. For more information on the API, see Consumer API to Create Invoice.